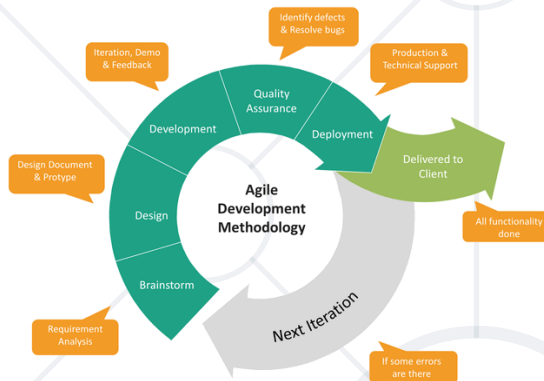


Software Development Methodologies

Software Project Management, Development Methodologies, Agile Development, Kanban, Scrum



SoftUni Team
Technical Trainers



SoftUni



Software University

<https://softuni.bg>

Table of Content

1. Software Project Management
2. Development Methodologies
3. Agile Development Principles
4. Kanban
5. Scrum
6. Scrumban
7. Project Tracking Tools





Software Project Management

Plan, Monitor and Control Project Development

What is Project Management?

- **Project management**
 - **Organizing** and **managing** work and resources in order to successfully complete a project
 - "Successfully" == within defined **scope**, **quality**, **time** and **cost** constraints
- **Project**
 - Temporary undertaking that aims to deliver some specific result
- **Project manager**
 - Person who plans, monitors and manages the project
 - Often this is the "**team leader**"



What is Software Project Management?

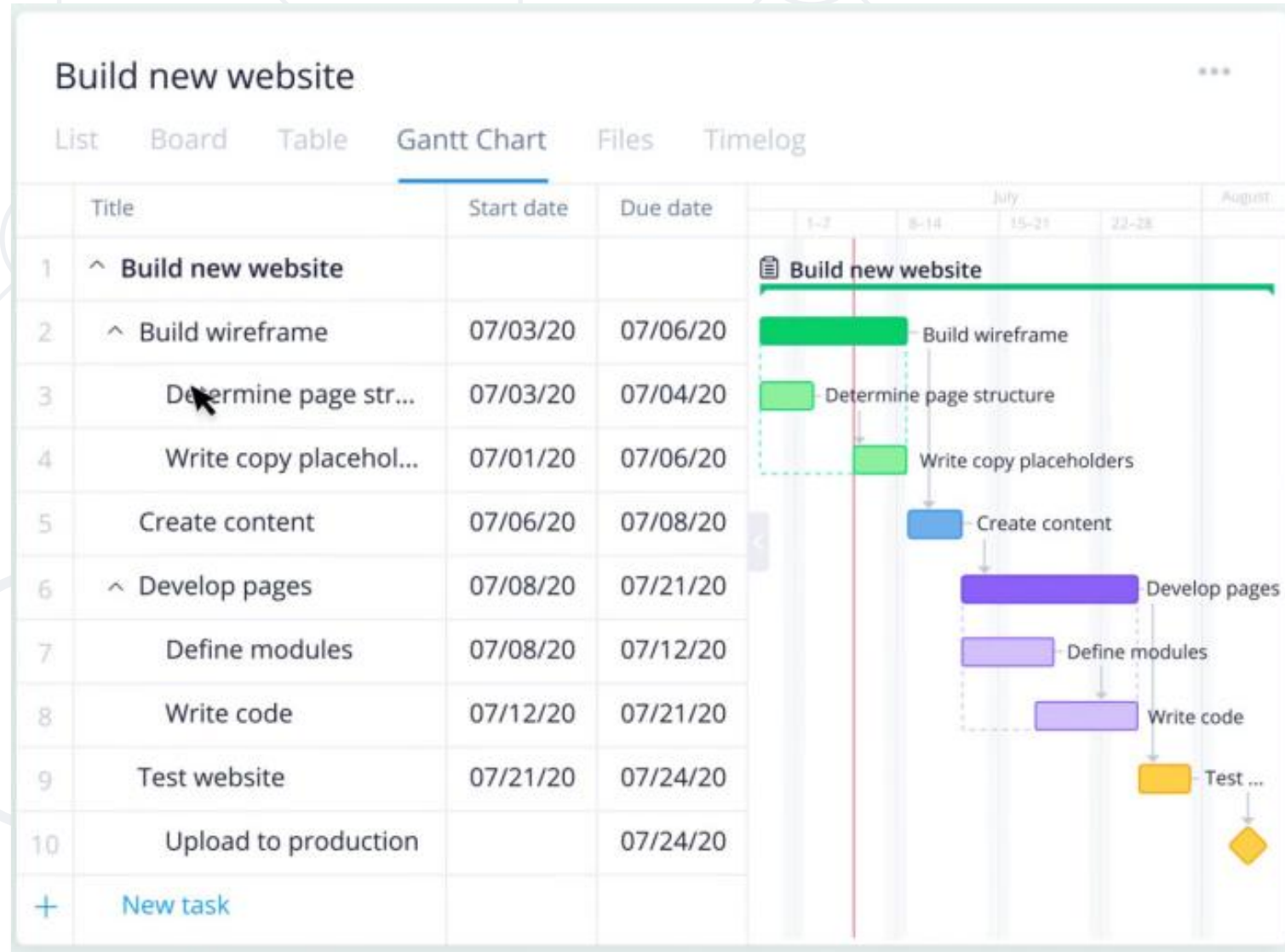
- **Software project management**
 - Management discipline about **planning**, **monitoring** and **controlling** software projects
- **Project planning**
 - Identify the scope, estimate the work, identify tasks and milestones, and create a schedule
- **Project monitoring and control**
 - Monitor the project progress, manage risks, discuss and address issues, problems and delays



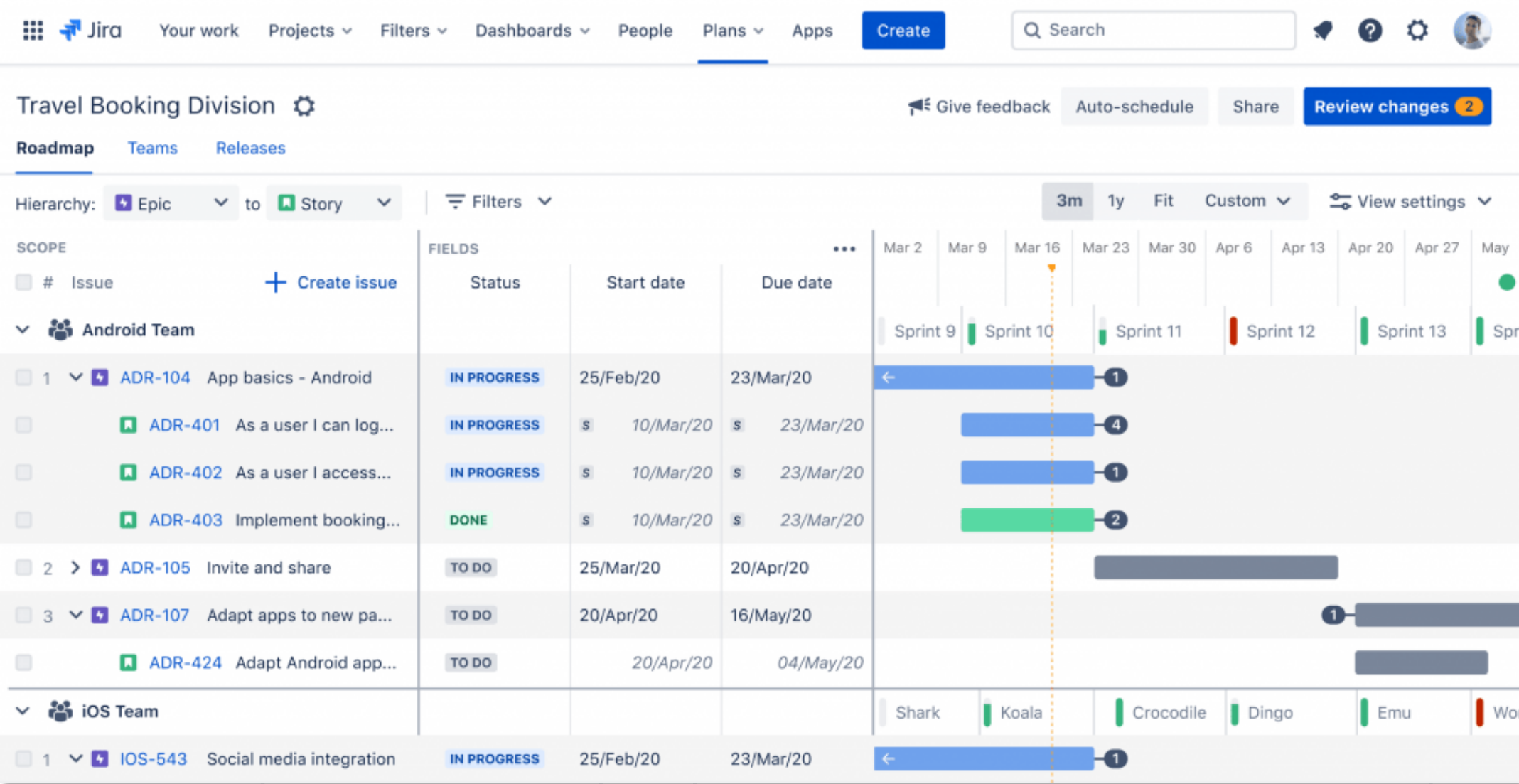
What is a Project Plan?

- The **project plan**
 - A document describing how project work will be organized
 - A chart / diagram / spreadsheet
 - Contains **tasks, resources, schedule, milestones**, progress indicators, etc.
 - Tasks may have **start, end, assigned resources** (team members), **% complete, dependencies, nested tasks, cost**, etc.
- **Project management tools** (like MS Project, Zoho Projects, Monday.com) simplify **creating** and **monitoring** project plans

Project Plan – Example



Project Plan – Example



- **Classical project management**
 - Create a plan and execute it
- **Agile project management**
 - Less planning and more flexibility
 - Changes are welcome
- **Task boards** visualize and organize the project tasks / people / progress
 - Typical columns: TODO, Doing, Done
- Examples: [Trello](#), [Asana](#), [GitHub Projects](#), [Azure Boards](#)

Task Board – Example

Done 4

❗ Add spread/rest types to support proposed object spread/rest
#10727 opened by sandersn
Fixed Suggestion

❗ Error in find-all-references for 3 spread types
#13654 opened by andy-ms
Bug Fixed

🚫 Implementing DeepMerge
#12769 opened by AlexGalays
Bug

❗ FATAL ERROR:
CALL_AND_RETRY_LAST Allocation

In Progress 5

❗ JSDoc: support intersection types
#16550 opened by OliverJAsh
Domain: JSDoc Salsa Suggestion

❗ Intellisense for object properties defined in multi-line JSDoc comments
#11597 opened by abgivant
Domain: JSDoc Salsa Suggestion
VS Code Tracked

❗ Support Object.<type1, type2> type notation in JSDoc
#15105 opened by mjbvz
Bug Domain: JSDoc Salsa
VS Code Tracked

Not started 17

❗ Inconsistent treatment of non-widening string literal types and `+=`
#13865 opened by mariusschulz
Bug

❗ Generics and keyof does not work with computed properties
#14473 opened by altschuler
Bug

❗ Type inference failure with discriminated type and partially overlapping type.
#16513 opened by yannack
Bug



Live Demo

Project Taskboard

<https://github.com/microsoft/TypeScript/projects/13>



Development Methodologies

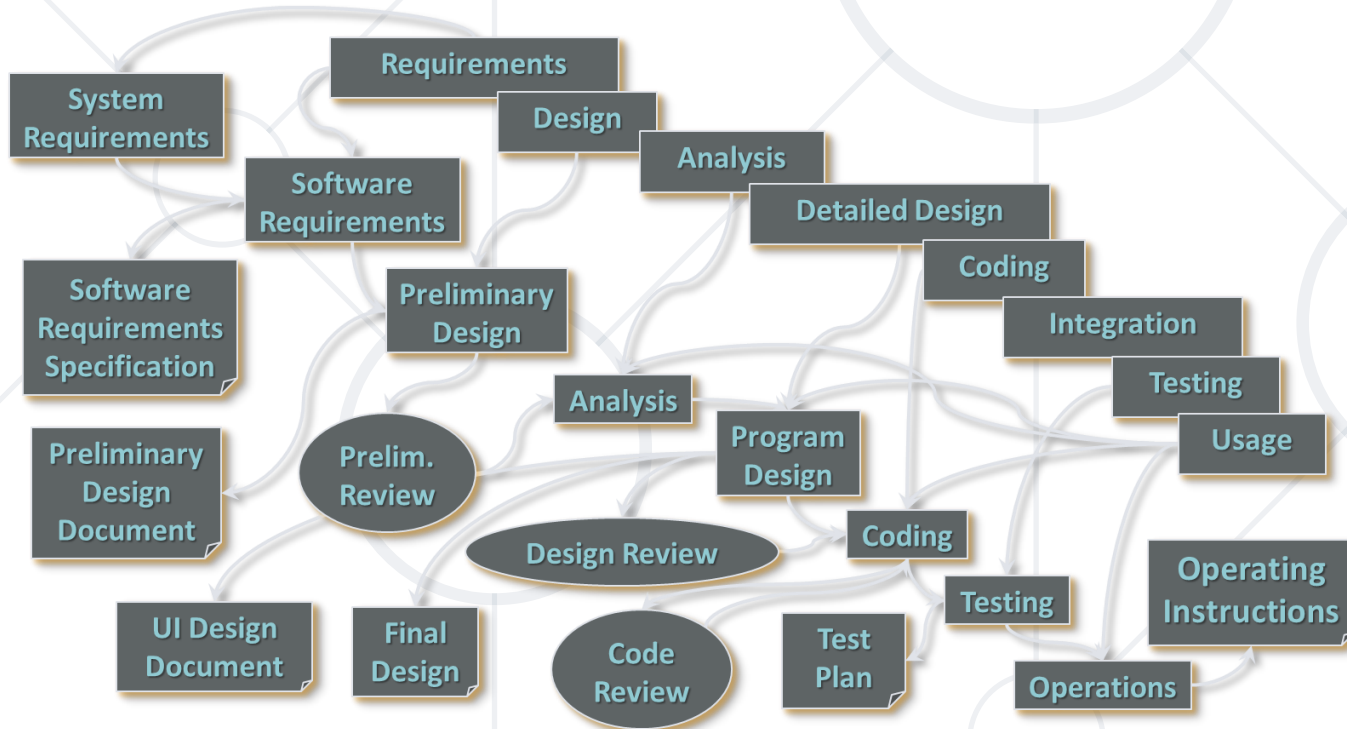
Waterfall, Scrum, Kanban, Extreme Programming

What is a Development Methodology?

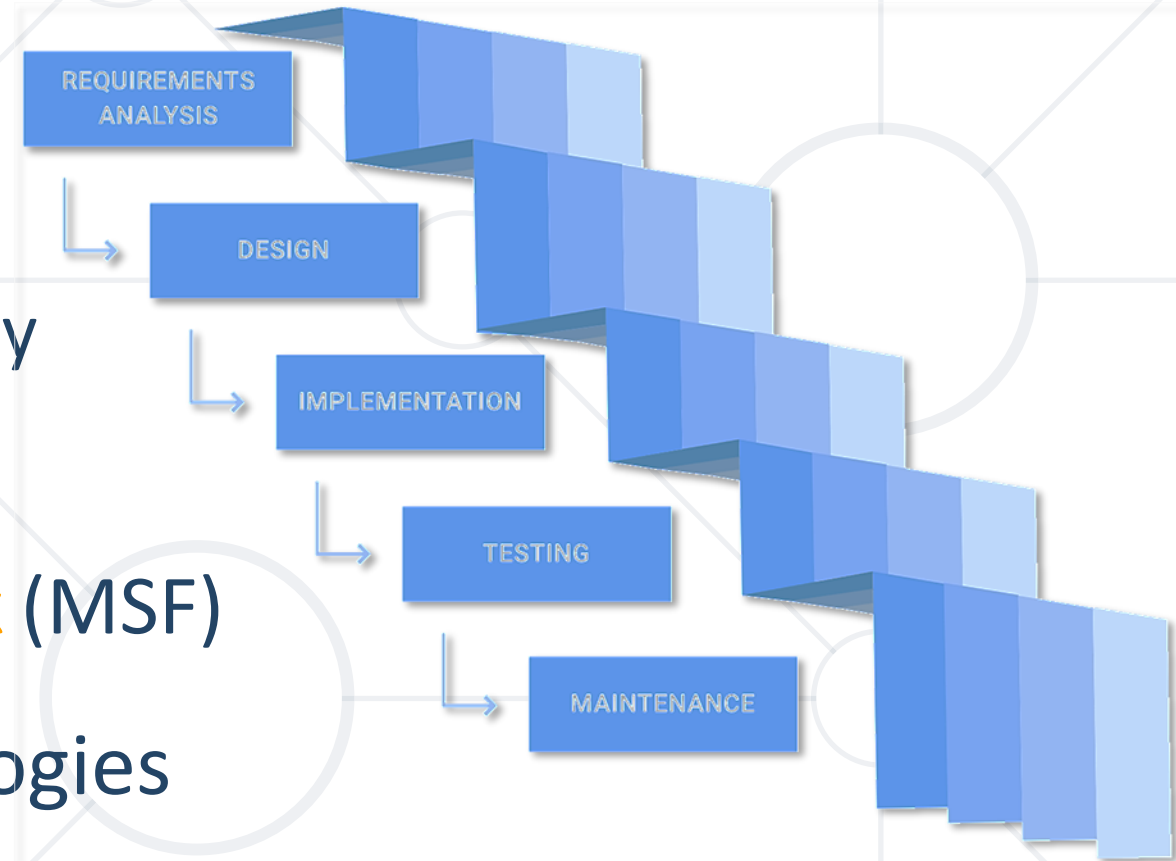
- A set of **practices** and **procedures** for organizing the **development process**
- A systematic, **engineering approach** for organizing and managing software projects
- A set of **principles, conventions, processes** and **procedures** the organization decides to follow
- A set of **rules** that developers must follow
- **Principles** and **processes** to define how to order tasks, perform monitoring, and control

Heavyweight and Agile Methodologies

- **Heavy methodologies** rely on **formal procedures** and **documents**
 - Lots of documents, diagrams, formal approvals, etc.
- **Agile methodologies** rely on **small iterations** and **less formalities**



- Back in history
 - The **"Waterfall"** process
 - Old-fashioned, rarely used today
 - **Rational Unified Process** (RUP)
 - **Microsoft Solutions Framework** (MSF)
- Modern development methodologies
 - **Agile** development processes
 - **Scrum, Kanban, Lean Development, Scrumban, Hybrid, ...**





Agile Development

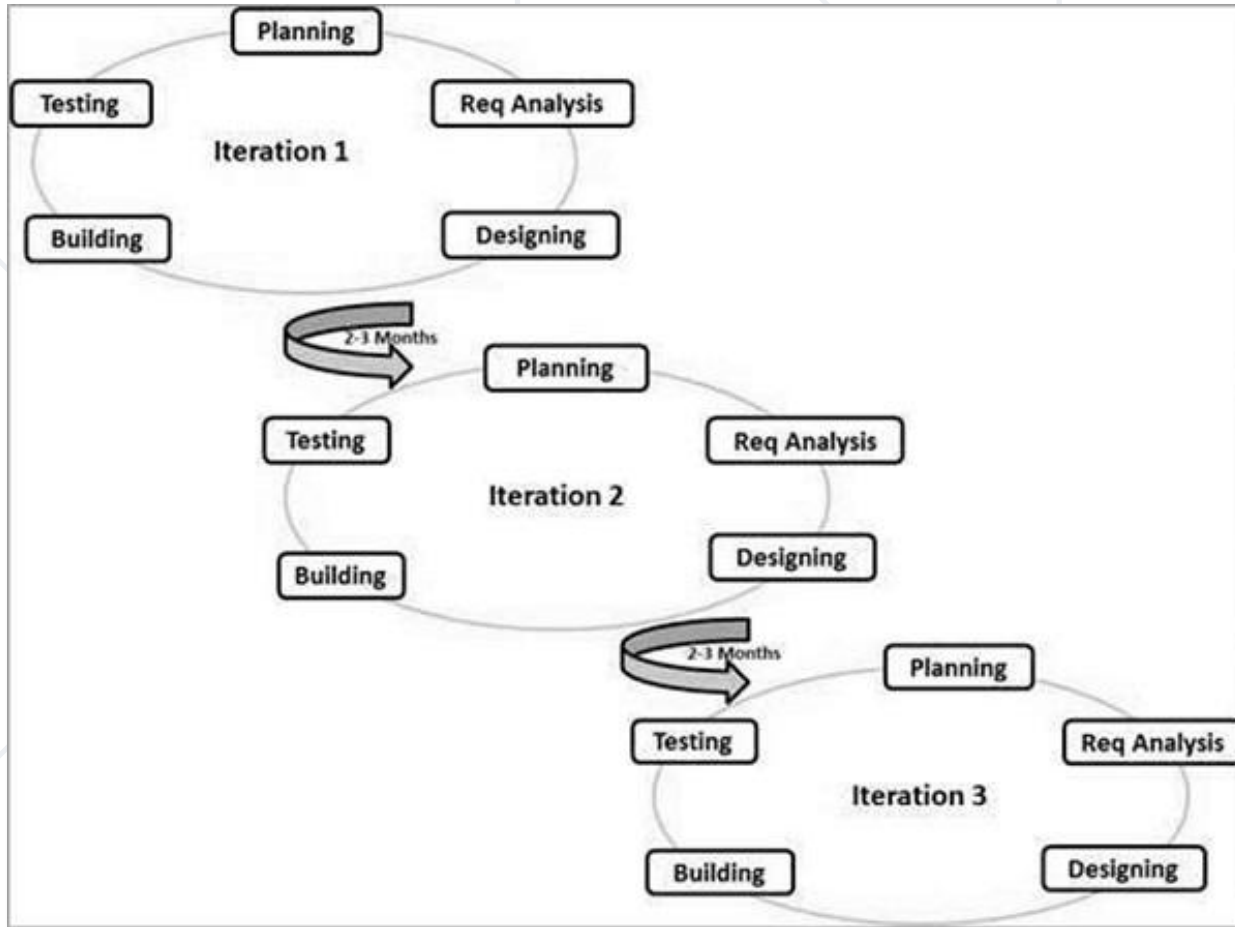
Lightweight, Incremental, Responding to Change

- The **agile manifesto** (Kent Beck, Martin Fowler, Robert Martin, ...)

"Our highest priority is to satisfy the customer through early and continuous delivery of valuable software"

- The **agile** spirit
 - **Incremental**: working software frequently over heavy documentation
 - **Cooperation**: customer collaboration over contract negotiation
 - **Straightforward**: individuals + interactions over processes + tools
 - **Adaptive**: responding to change over following a plan

Iterative and Incremental Development



- Agile development relies on **increments** (steps)
 - Small portions of software, developed in iterations
- Each **iteration** goes through own dev lifecycle
 - Requirements → design → code → test → deploy

- **Scrum**: sprint based, biggest market share
- **Kanban**: task board based
- **Scrumban**: Scrum + Kanban
- **Lean** Software Development
- **XP** (eXtreme Programming)
- **Hybrid** (multiple methods)



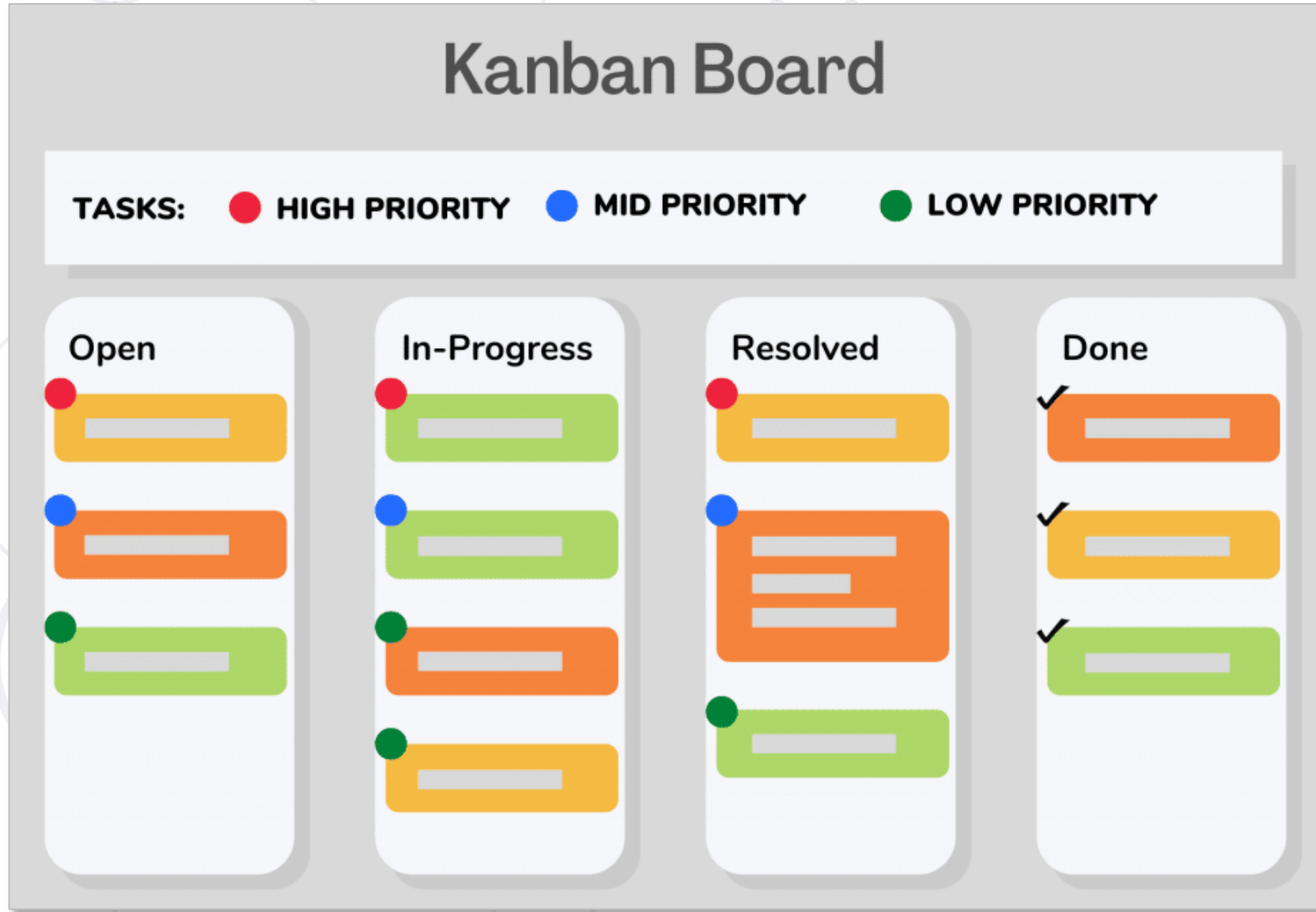


Kanban

Kanban, Principles, Tasks, Flow, Kanban Boards

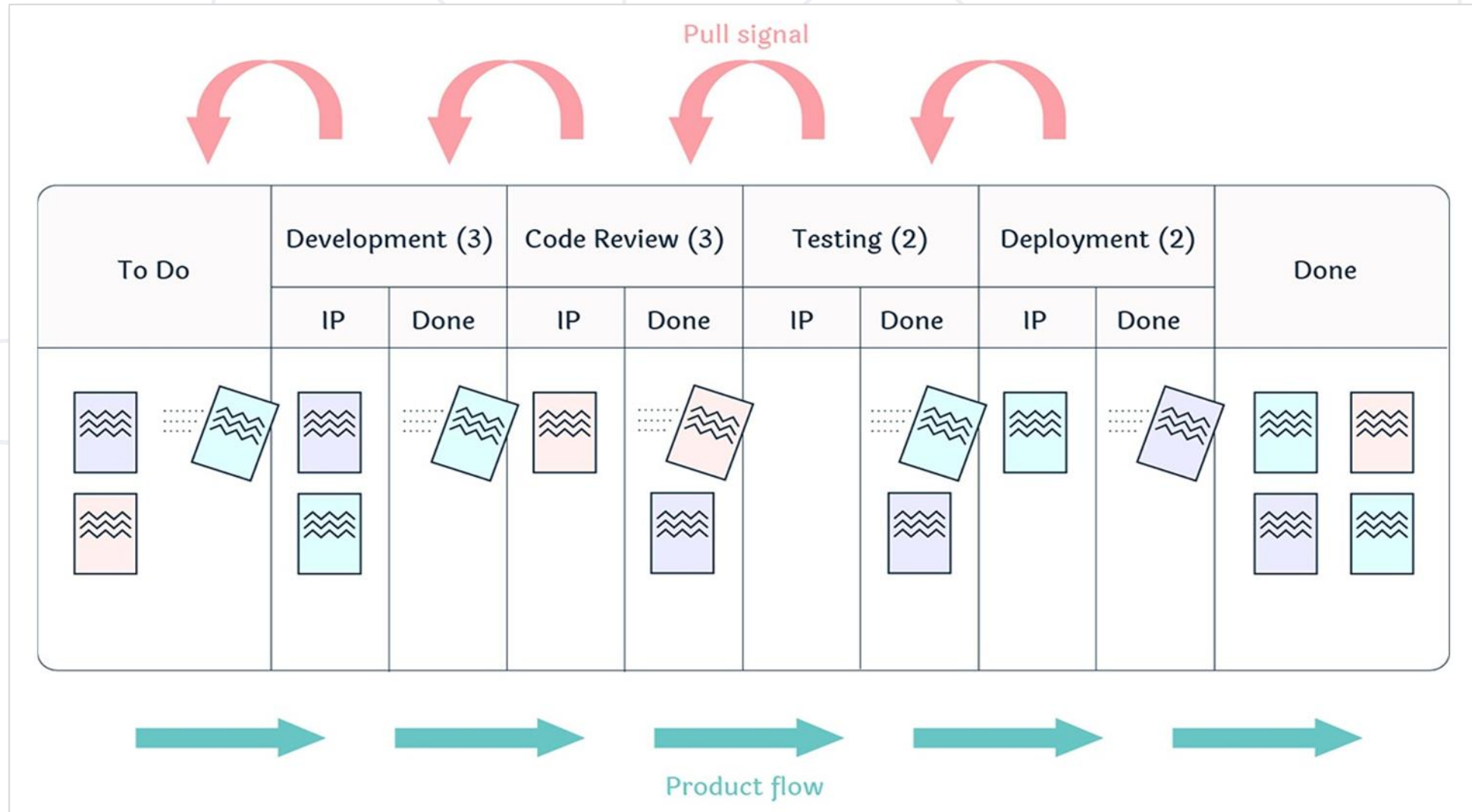
- **Kanban**
 - Popular agile project management framework
- **Visualization**
 - Shared Kanban board
- **Flow of tasks**
 - Open → In-Progress → Done
- **Limiting work in progress**
 - Minimize multi-tasking
- **Continuous improvement**
 - Monitor, analyze, improve

Kanban Board: Structure

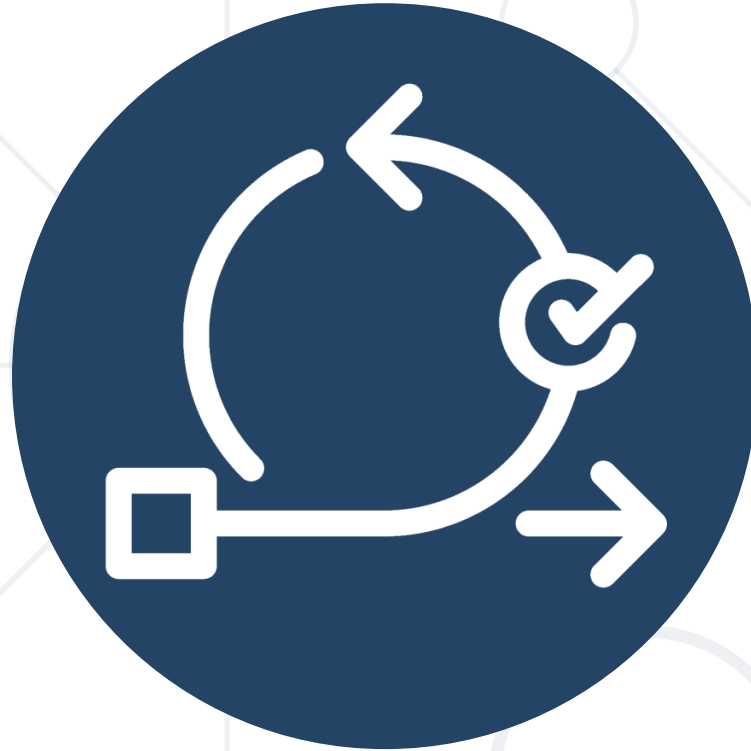


- Start with **what you do** now
 - Respect the **current process**, roles, responsibilities, job titles
- **Visualize the workflow**
 - Use Kanban board
- **Manage workflow**
 - Fast smooth flow
 - **Limit the work-in-progress** (WIP) items
- **Incremental**, evolutionary change

Kanban in Software Development



- Most **Kanban boards** include the following **stages**
 - **Waiting**
 - Typically **includes** the **backlog** of tasks that are waiting to be **opened up**
 - **In-progress**
 - **Holds all** the **tasks** that are **currently being worked on**
 - **Completed**
 - Holds **done** tasks
 - **Blocked**
 - When a task **can't** be **completed** or its progress is **halted**, **until** it **can** be **picked up again**



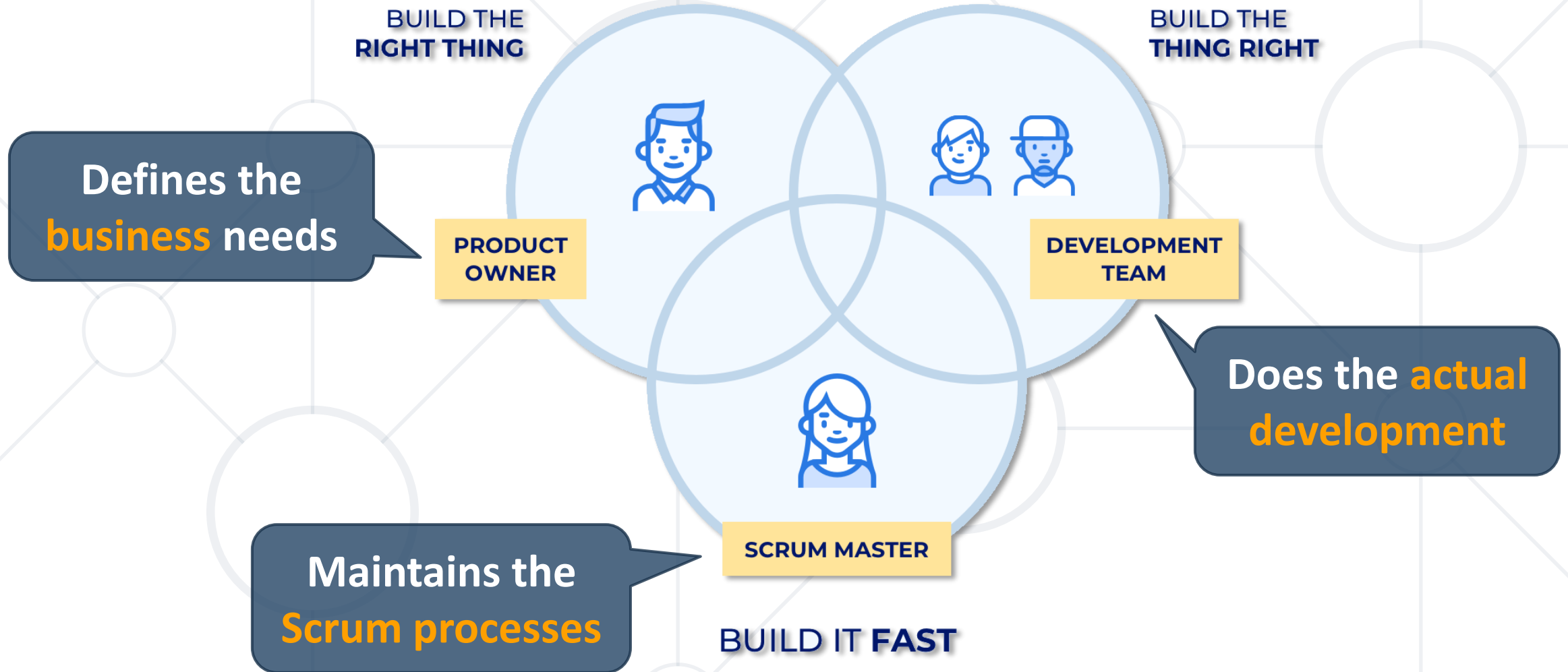
Scrum

Backlog, Sprints, Daily Scrum, Scrum Roles

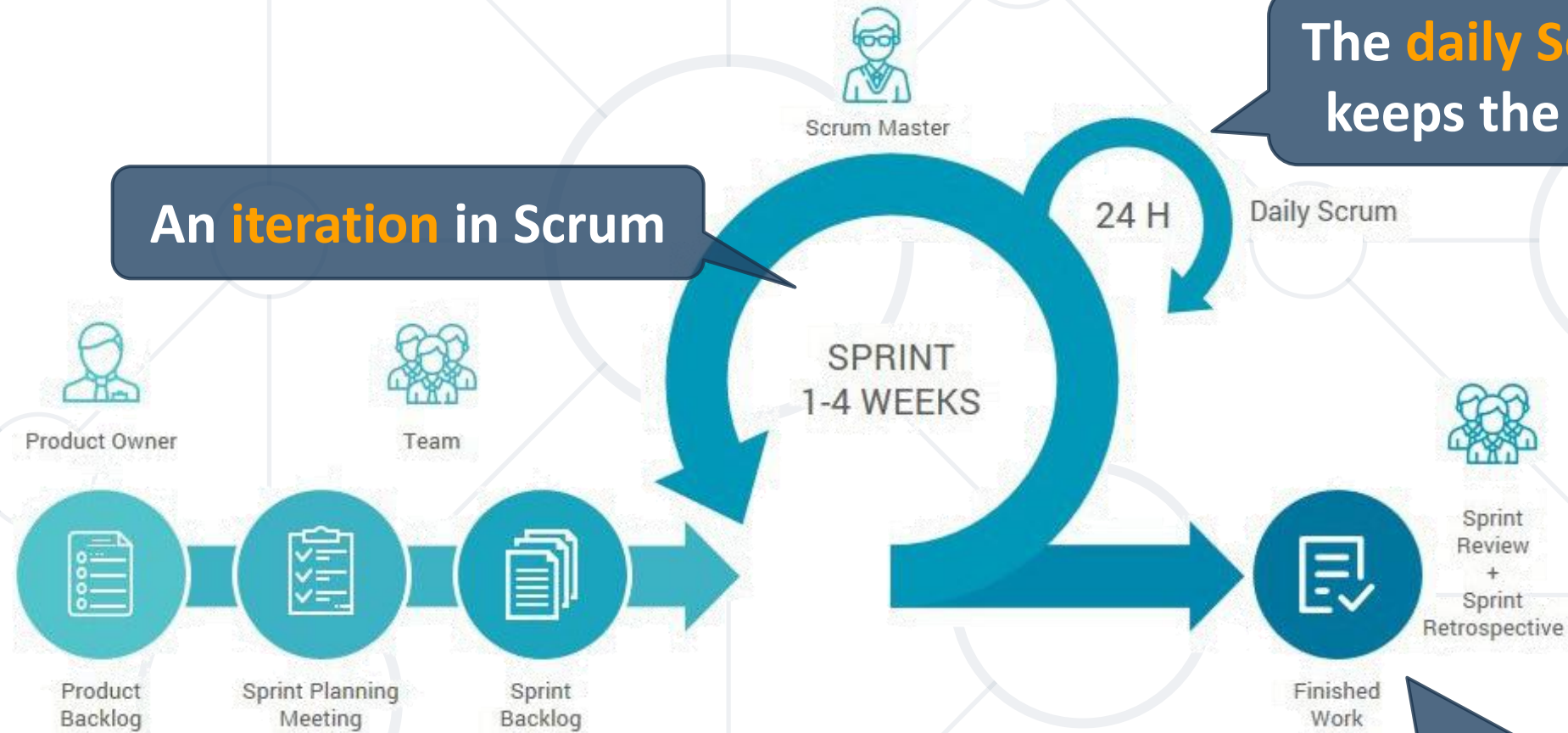
■ Scrum

- An iterative and incremental **agile software development methodology** for product development
- Highly popular in the software industry (50%+ market share)
- Product features (user stories) stay in the product "**backlog**"
- Software is developed in iterations, called "**sprints**"
 - At each sprint, a **set of user stories** are implemented
 - A sprint lasts $\approx$ **1-4 weeks**
 - At the end, the team delivers fully **working features**

Scrum Roles



Scrum Terminology



An **iteration** in Scrum

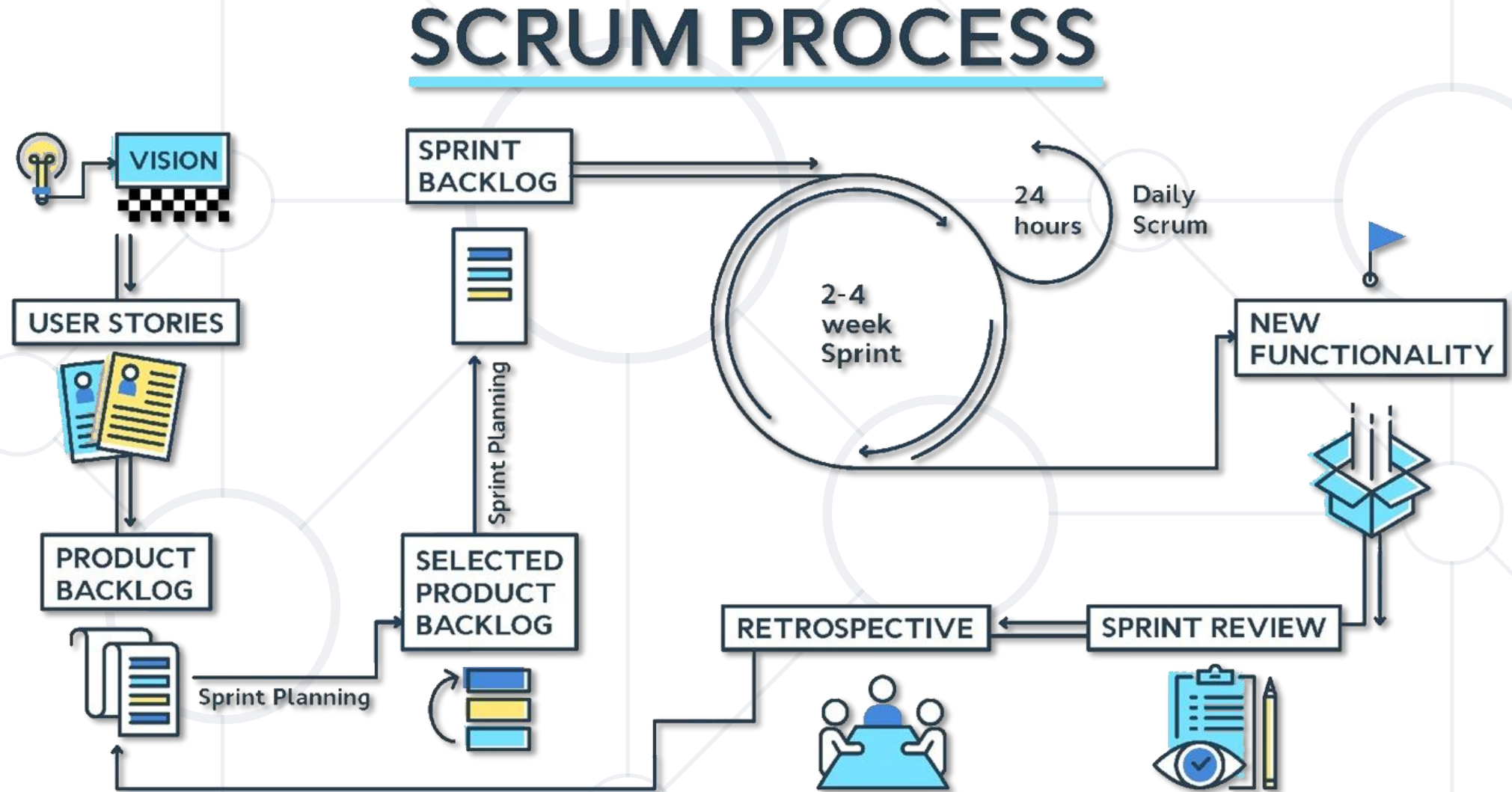
The **daily Scrum meeting** keeps the team in sync

All **features** that have to be **developed**

All **features** planned for the **current sprint**

Sprints end with **working features + review meeting**

The Scrum Process Framework



- **Sprint Planning Meeting**
 - At the beginning of the sprint cycle
 - Establish the Sprint backlog
- **Daily Scrum stand-up meeting**
 - Each day during the sprint
 - Project status from each team member
 - Timeboxed to 15 minutes
- **Sprint Review Meeting**
 - Review the work completed / not completed





Scrumban

Hybrid between Scrum and Kanban

- A hybrid between **Scrum** and **Kanban**
 - Combines the **structure of Scrum** with the **continuous flow** of work, **flexibility** and **visualization of Kanban**
- Work performed in **short iterations**
- **Limited number** of work-in-progress items
 - Monitored and tracked with Kanban boards
- No sprints
 - Work can be introduced and prioritized at any time
- No team roles
 - More flexibility

Differences Between Kanban and Scrum

	Kanban	Scrum
Roles	Roles are fluid . Project manager optional.	Roles are predefined . Scrum master required.
Tasks	Tasks are shared by everyone.	Tasks have assigned owners.
Timeline	Timelines evolve on an as needed basis (more flexible).	Timelines are timeboxed into sprints.
Changes	Changes in work items and their priorities can be made at any time .	Changes can only be made between sprints (upon completion of a sprint).
Productivity	Productivity is measured by the cycle time of the complete project.	Productivity is measured by the story points completed in each sprint.



Project Tracking Tools

Task Management Tools/ Issue Tracker Systems

- **GitHub Projects**
 - A free Kanban board in GitHub
 - Microsoft example



VSCode Git/SCM

General GitHub

-status: "✓ Done" 17 X

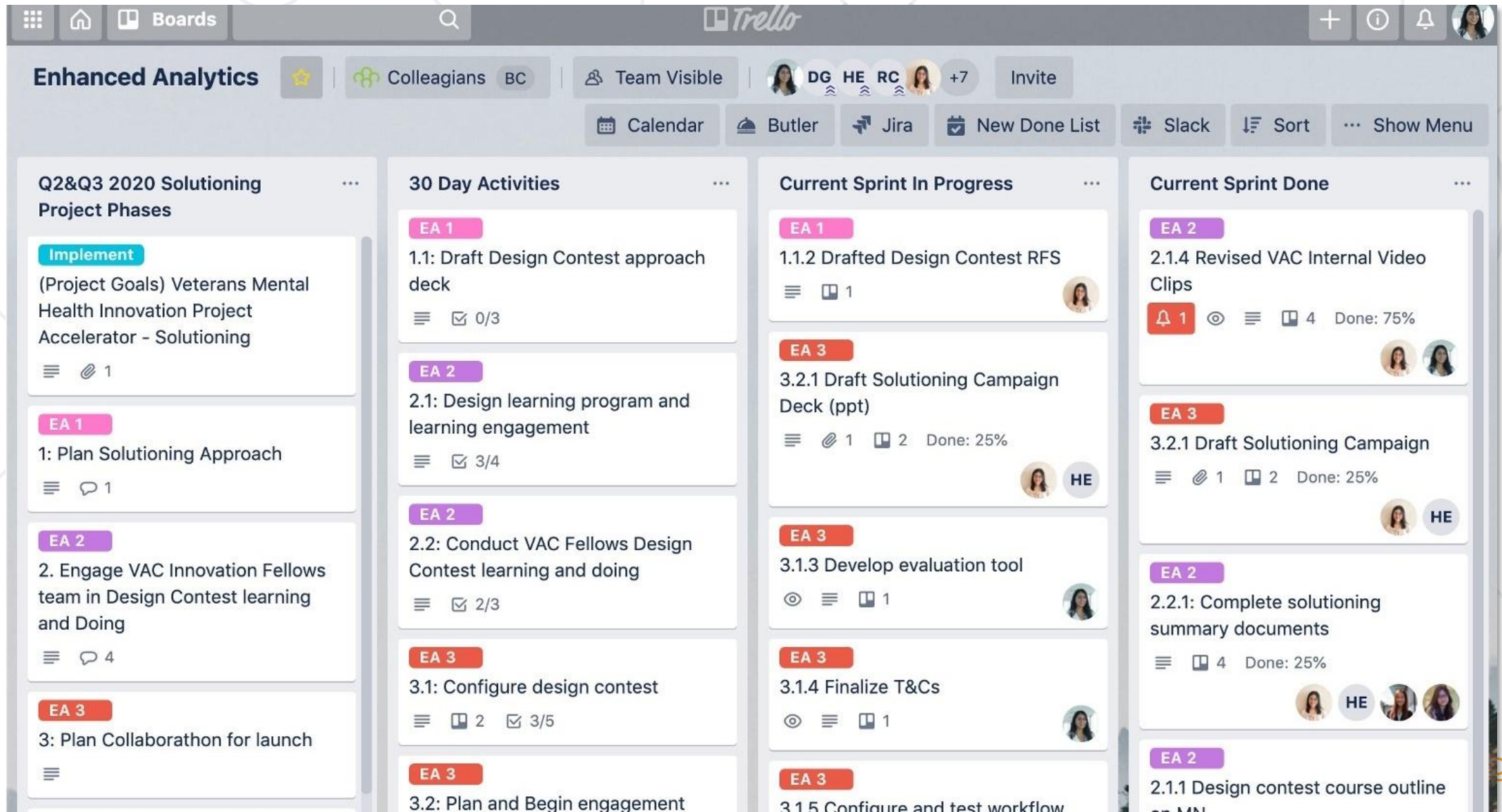
	Title	...	Assignees	...
23	Git: Prevent running git operations during sync #60630		joaomoreno and lszomoru	
24	Git: Sync Changes Button Should Not Pull When There Are No Incoming Changes #163934		lszomoru	
25	SCM title menu Create Pull Request action is unavailable when in a Remote window #3911		alex00 and lszomoru	
26	Git error (Cannot fast-forward to multiple branches.) when syncing during autofetch #158309		lszomoru	
29	[Bug] editor could not be opened if a file have conflict with added by us #153639		lszomoru	
31	VS Code does not respect the diff flag in .gitattributes #154815		lszomoru	
33	After VSCode is updated, git@github.com: Permission denied (publickey). #149622		lszomoru	

Trello

- Trello
 - The **visual tool** that empowers your team to **manage** any **type** of **project**, **workflow**, or **task tracking**
 - Trello's look and feel are based on the **principles** of a **Kanban board**
 - **Visual tool** made up of **cards** and **columns** to **track progress**
 - Think of **Trello** as a **digital, living whiteboard** in **motion**



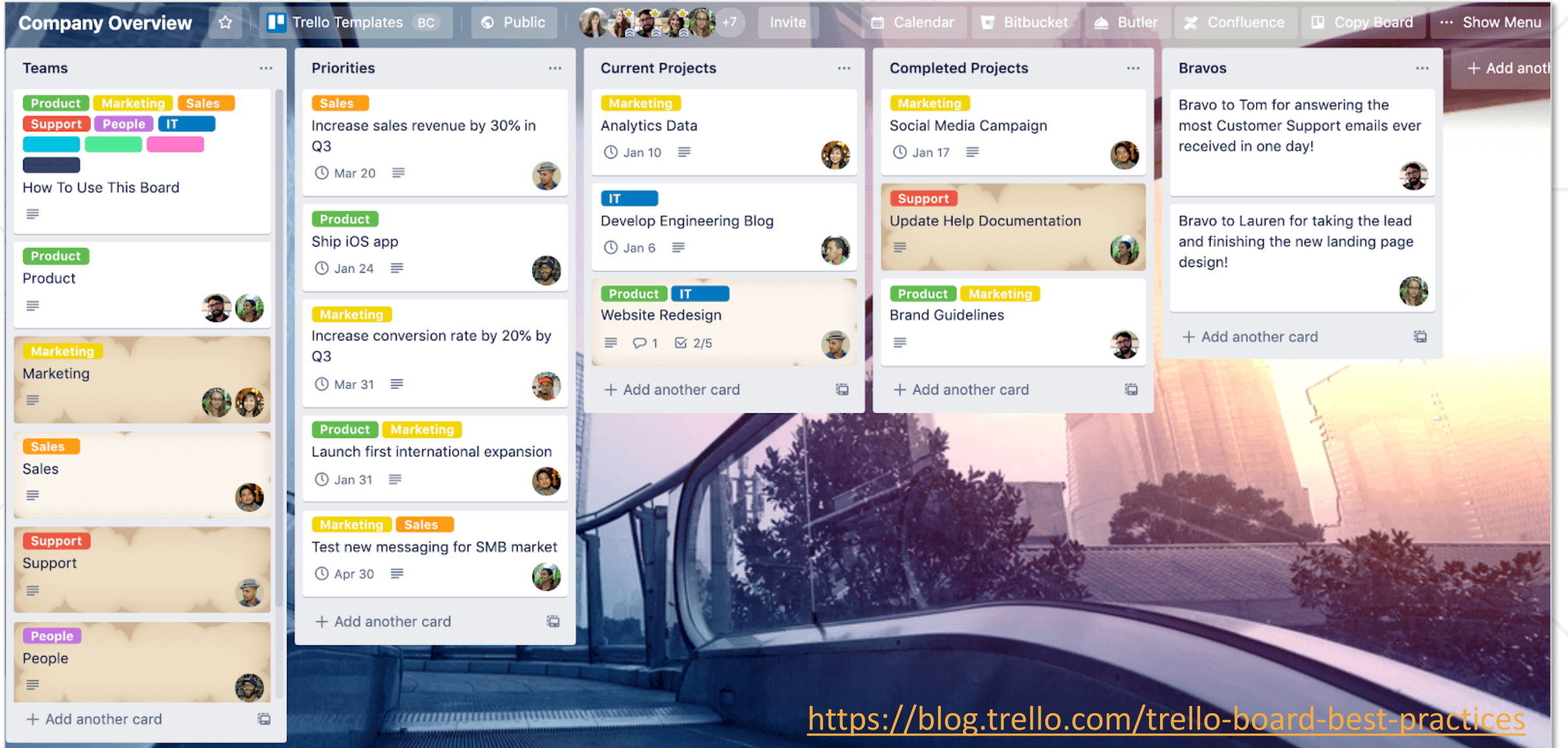
Trello – A Simple Kanban Board



The screenshot displays a Trello workspace with the following components:

- Navigation Bar:** Includes icons for Boards, Search, and user profile. The workspace name "Colleagians BC" is visible.
- Team & Tools:** Shows "Team Visible" with member avatars (DG, HE, RC, +7) and an "Invite" button. Below this are buttons for "Calendar", "Butler", "Jira", "New Done List", "Slack", "Sort", and "Show Menu".
- Columns and Cards:**
 - Q2&Q3 2020 Solutioning Project Phases:**
 - Implement** (blue label): (Project Goals) Veterans Mental Health Innovation Project Accelerator - Solutioning. 1 comment.
 - EA 1** (pink label): 1: Plan Solutioning Approach. 1 comment.
 - EA 2** (purple label): 2. Engage VAC Innovation Fellows team in Design Contest learning and Doing. 4 comments.
 - EA 3** (red label): 3: Plan Collaborathon for launch.
 - 30 Day Activities:**
 - EA 1** (pink label): 1.1: Draft Design Contest approach deck. 0/3 checklist items.
 - EA 2** (purple label): 2.1: Design learning program and learning engagement. 3/4 checklist items.
 - EA 2** (purple label): 2.2: Conduct VAC Fellows Design Contest learning and doing. 2/3 checklist items.
 - EA 3** (red label): 3.1: Configure design contest. 2 attachments, 3/5 checklist items.
 - EA 3** (red label): 3.2: Plan and Begin engagement.
 - Current Sprint In Progress:**
 - EA 1** (pink label): 1.1.2 Drafted Design Contest RFS. 1 attachment.
 - EA 3** (red label): 3.2.1 Draft Solutioning Campaign Deck (ppt). 1 attachment, 2 attachments, Done: 25%.
 - EA 3** (red label): 3.1.3 Develop evaluation tool. 1 attachment.
 - EA 3** (red label): 3.1.4 Finalize T&Cs. 1 attachment.
 - EA 3** (red label): 3.1.5 Configure and test workflow.
 - Current Sprint Done:**
 - EA 2** (purple label): 2.1.4 Revised VAC Internal Video Clips. 1 notification, 4 attachments, Done: 75%.
 - EA 3** (red label): 3.2.1 Draft Solutioning Campaign. 1 attachment, 2 attachments, Done: 25%.
 - EA 2** (purple label): 2.2.1: Complete solutioning summary documents. 4 attachments, Done: 25%.
 - EA 2** (purple label): 2.1.1 Design contest course outline.

Trello – A Simple Kanban Board



The screenshot displays a Trello board titled "Company Overview" with a top navigation bar including "Trello Templates", "Public", "Invite", "Calendar", "Bitbucket", "Butler", "Confluence", "Copy Board", and "Show Menu". The board is organized into five main columns:

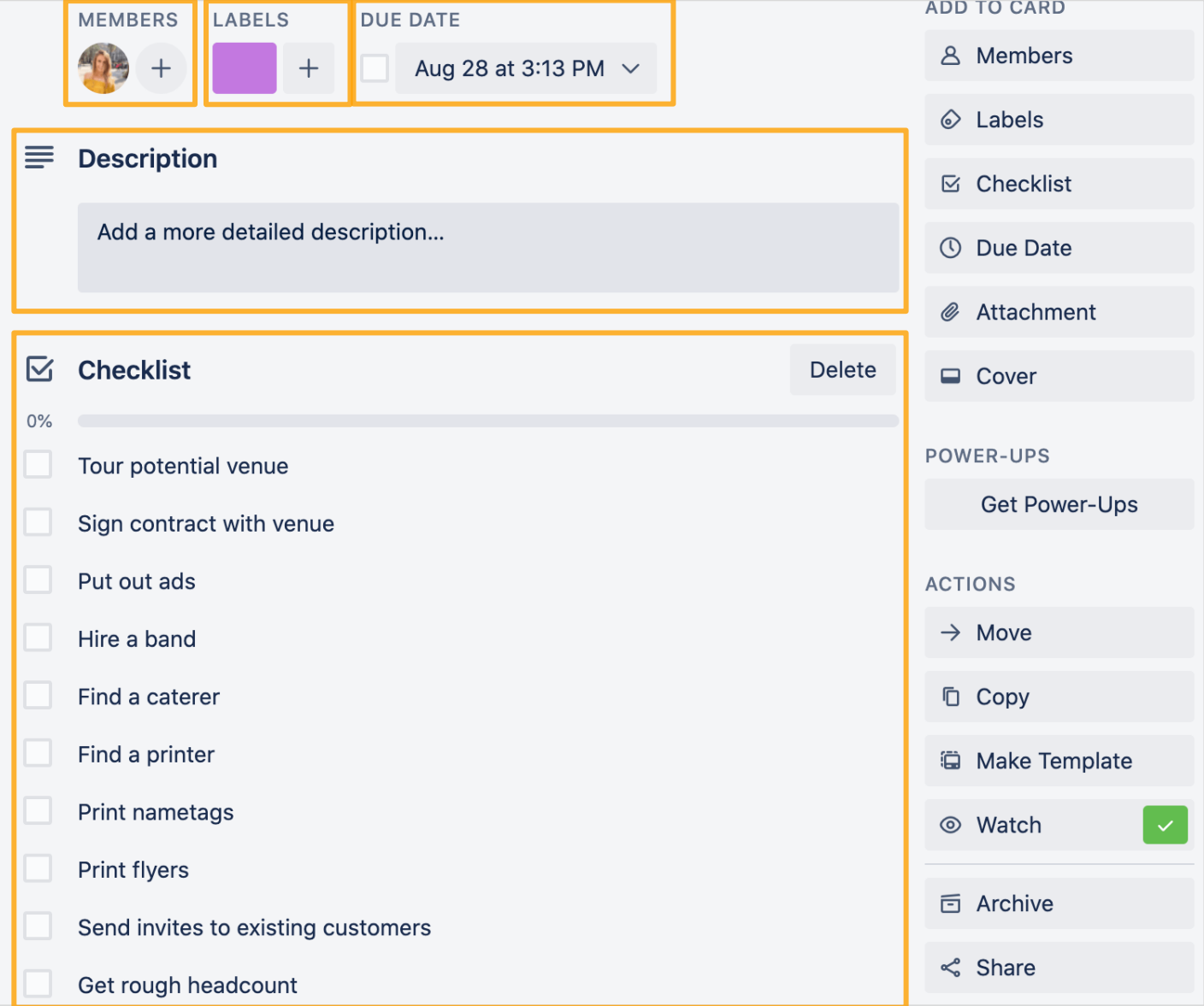
- Teams:** A list of team categories including Product, Marketing, Sales, Support, People, and IT, each with a corresponding colored card and a "How To Use This Board" link.
- Priorities:** A list of tasks categorized by priority (Sales, Product, Marketing, Product Marketing, Marketing Sales) with due dates and assignees.
- Current Projects:** A list of ongoing projects (Analytics Data, Develop Engineering Blog, Website Redesign) with due dates and assignees.
- Completed Projects:** A list of completed projects (Social Media Campaign, Update Help Documentation, Brand Guidelines) with completion dates and assignees.
- Bravos:** A list of celebratory messages (Bravo to Tom for answering the most Customer Support emails ever received in one day!, Bravo to Lauren for taking the lead and finishing the new landing page design!) with assignees.

Each column has a "+ Add another card" button at the bottom. The background of the board is a scenic view of a city skyline at sunset.

<https://blog.trello.com/trello-board-best-practices>

Trello – Example

- A card may have
 - Title
 - Members
 - Labels
 - Due date
 - Description
 - Attachments
 - Tasks checklist
 - Activity
 - Comments



The screenshot shows a Trello card interface. Orange boxes highlight the following elements:

- MEMBERS:** A box containing a member's profile picture and a plus sign to add more members.
- LABELS:** A box containing a purple label and a plus sign to add more labels.
- DUE DATE:** A box containing a date and time selector set to "Aug 28 at 3:13 PM" with a dropdown arrow.
- Description:** A large text area with a placeholder "Add a more detailed description..." and a hamburger menu icon on the left.
- Checklist:** A section titled "Checklist" with a "Delete" button. It includes a progress bar at 0% and a list of tasks with checkboxes:
 - ☐ Tour potential venue
 - ☐ Sign contract with venue
 - ☐ Put out ads
 - ☐ Hire a band
 - ☐ Find a caterer
 - ☐ Find a printer
 - ☐ Print nametags
 - ☐ Print flyers
 - ☐ Send invites to existing customers
 - ☐ Get rough headcount

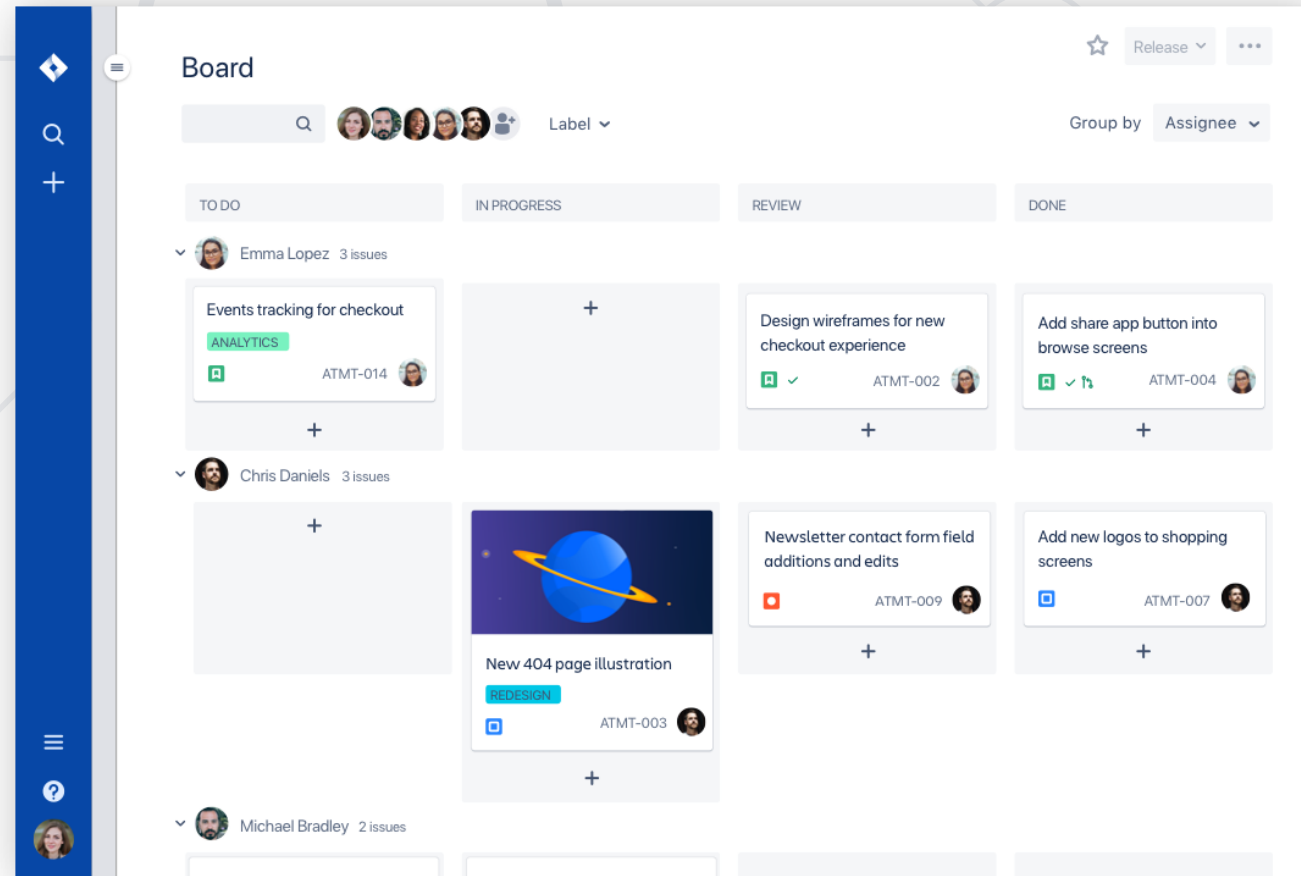
On the right side, the "ADD TO CARD" menu is visible, listing options: Members, Labels, Checklist, Due Date, Attachment, and Cover. Below this is the "POWER-UPS" section with a "Get Power-Ups" button. The "ACTIONS" section includes: Move, Copy, Make Template, Watch (which is selected with a green checkmark), Archive, and Share.

- A tool that allows agents to manage various cases and track their progress during the resolution process, beginning to end

- Tools

- Jira

- GitHub Issues



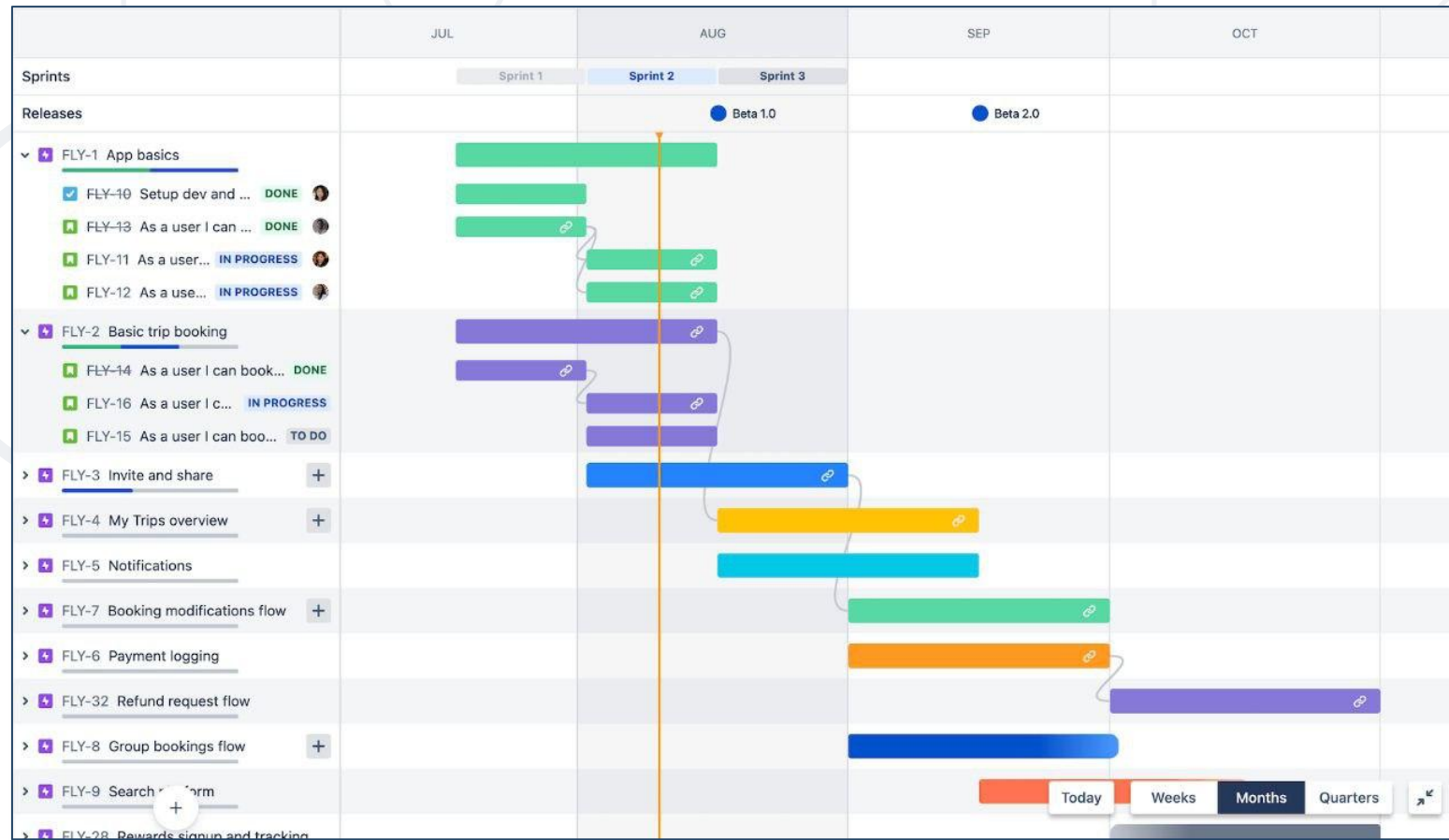
- 
- Widely used **project management** and **issue-tracking tool**
 - Primarily designed to **help teams plan, track, and manage their work efficiently**
 - Provides a **robust set of features** and **capabilities**



Jira Software

Key features of Jira

- Issue Tracking
- Project Management
- Customization and Extensibility
- Collaboration and Communication
- Integration Ecosystem
- Scalability and Deployment Options



- **Project**

- A collection of issues and related information

- **Issue**

- Bugs, features, user stories, tasks, or any other form of work that requires attention and action

- **Work Item**

- Work item, task, or a unit of work that needs to be tracked, managed, and resolved

■ Workflow

- Series of steps or statuses through which an issue progresses during its lifecycle
- Typically, it consists of a set of statuses (e.g., Open, In Progress, Resolved, Closed) and transitions between them

Sample Jira Issue

Jira

Dashboards

Projects

Issues

Boards

Portfolio

Power Apps

Insight

Assets

Create

Search

SCRUM / SCRUM-4

Copy&Sync Demo

Edit

Comment

Assign

More

To Do

Done

submmited

Admin

Details

Type: Task

Priority: Medium

Affects Version/s: None

Component/s: Demo

Labels: SCRUM-Sprint-1

Sprint: SCRUM Sprint 1

Description

Copy & Sync demo

Elements Checklist

Attachments

Available Operations

Issue Links

Status: TO DO

Resolution: Unresolved

Fix Version/s: 2.1.0

Dates

Agile

People

Assignee: Unassigned

Reporter: Admin

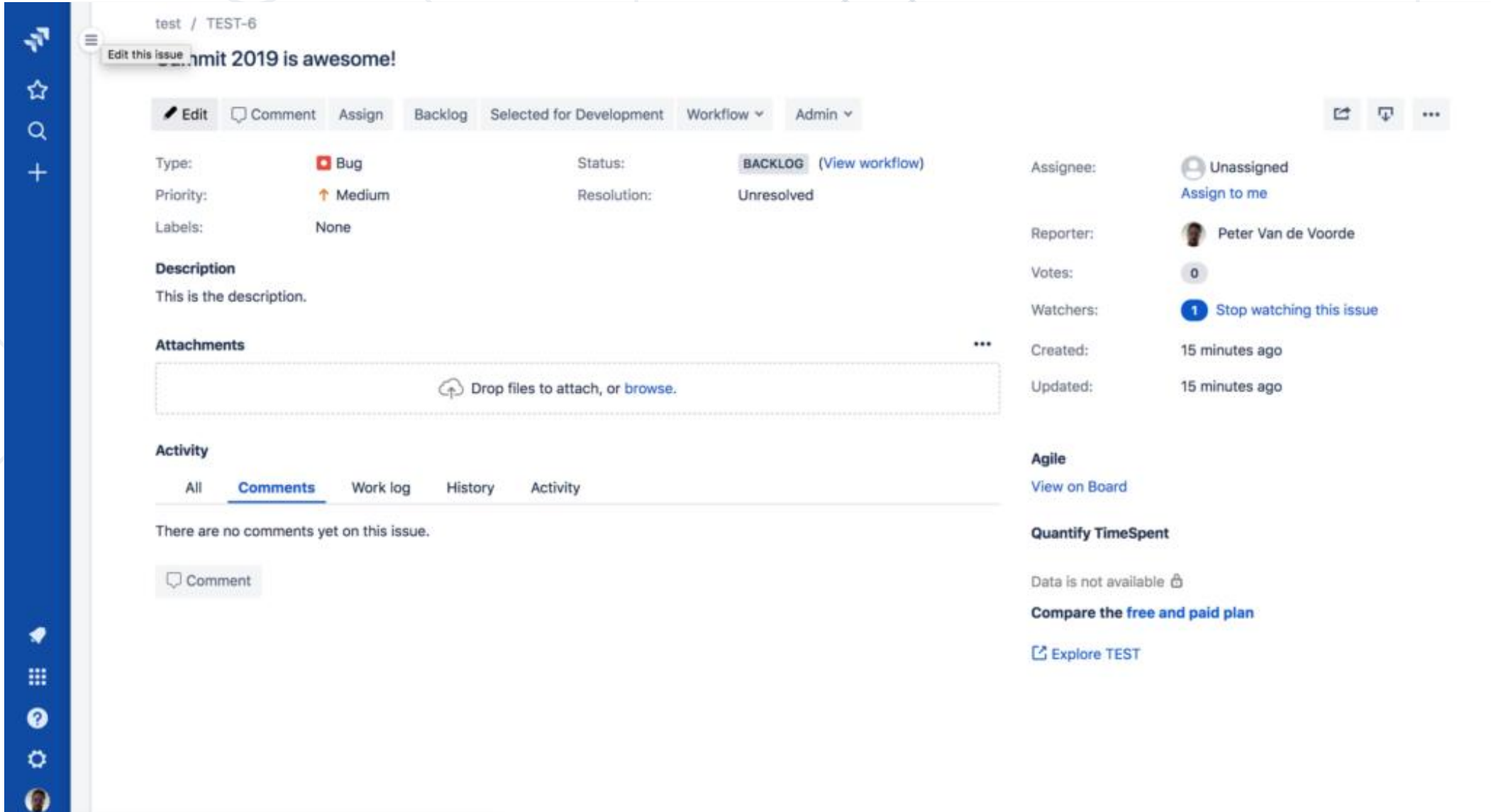
Votes: 0

Watchers: 1

Future Sprint: SCRUM Sprint 1

Wiki Page

Sample Jira Issue



test / TEST-6

Edit this issue Submit 2019 is awesome!

Edit Comment Assign Backlog Selected for Development Workflow Admin

Type: Bug Status: BACKLOG (View workflow)
Priority: Medium Resolution: Unresolved
Labels: None

Assignee: Unassigned
[Assign to me](#)

Reporter: Peter Van de Voorde

Votes: 0

Watchers: 1 [Stop watching this issue](#)

Created: 15 minutes ago
Updated: 15 minutes ago

Description
This is the description.

Attachments

Drop files to attach, or [browse](#).

Activity

All Comments Work log History Activity

There are no comments yet on this issue.

[Comment](#)

Agile
[View on Board](#)

Quantify TimeSpent
Data is not available 

[Compare the free and paid plan](#)

[Explore TEST](#)

- Software **project management** == **organizing** and **managing** work
- Development **methodology** == **heavyweight** and **agile**
- **Agile** development == relies on **increments**, each iteration goes through its **own lifecycle**
- **Kanban** == popular **agile** project **management framework**
- **Scrum** == **agile** software **development methodology** for product development
- **Scrumban** == **hybrid** between **Scrum** and **Kanban**
- Use **task management tools** (Trello, GitHub Projects, Asana)
- Use **issue trackers** for bugs/tickets (GitHub Issues, JIRA)



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